Multi-UAV Source Seeking and Optimization

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Abstract

This report summarizes the work conducted for the BTP Project, focusing on Multi-UAV Source Seeking and Optimization. We investigate the performance of different Particle Swarm Optimization (PSO) variants from [2], including Standard PSO (SPSO), Acceleration-based PSO (APSO), and Adaptive Random PSO (ARPSO), in source seeking tasks under varying noise levels. Furthermore, we evaluate these algorithms on the CEC 2022 benchmark suite to assess their optimization capabilities across different dimensions (2D, 10D, 20D). We introduce and analyze hybrid strategies, Hybrid Parallel PSO and Hybrid Sequential PSO, to combine the strengths of exploration and exploitation. Our findings demonstrate that while APSO excels in moderate dimensions and noisy source seeking, the Hybrid Sequential strategy offers superior performance in high-dimensional optimization problems, highlighting the importance of a phased exploration-exploitation approach.

1 Introduction

Building upon previous work which focused on understanding different PSO variants and implementing exploration-based source seeking, this phase of the project targets exploitation strategies. The primary objectives are:

- To try exploitation strategies in source seeking algorithms and observe the behaviors of different PSOs.
- To identify which PSO or combination of different PSOs work best in specific scenarios.
- To develop and evaluate hybrid Particle Swarm Optimization approaches that combine the strengths of different PSO variants.
- To analyze how different hybridization strategies affect convergence behavior and solution quality.

2 Particle Swarm Optimization (PSO) Variants

PSO is a technique used to find an approximate solution for maximization or minimization problems. It initializes with a group of random particles and searches for the optimal solution by updating generations. Particles move through the solution space, evaluated by a fitness criterion. Each particle updates its position based on its personal best (P_{best}) and the global best (g_{best}) of the swarm.

2.1 Standard PSO (SPSO)

SPSO uses a direct velocity-based update mechanism (first-order dynamics) and is a widely used baseline. It offers a balanced exploration-exploitation trade-off.

$$v_i(t+1) = \omega v_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i) \tag{1}$$
$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{2}$$

2.2 Acceleration-based PSO (APSO)

APSO uses acceleration as the primary update mechanism (second-order dynamics), maintaining momentum through previous movement history. This produces smoother trajectories.

$$a_i(t+1) = \omega_1 a_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i) \tag{3}$$
$$v_i(t+1) = \omega_2 v_i(t) + a_i(t+1)T \tag{4}$$
$$x_i(t+1) = x_i(t) + v_i(t+1)T \tag{5}$$

2.3 Adaptive Random PSO (ARPSO)

ARPSO incorporates adaptive parameters and randomness, dynamically adjusting inertia weight based on swarm state and including additional random attractive positions for exploration.

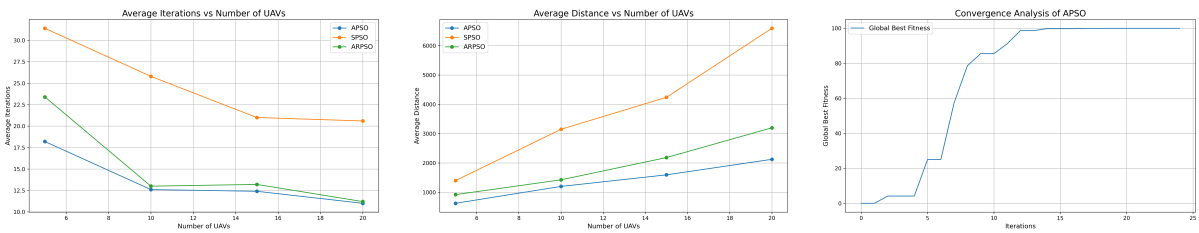
$$\omega_i(t+1) = f(\text{evolutionary speed, aggregation degree}) \tag{6}$$
$$v_i(t+1) = \omega_i v_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i) + c_3 r_3 (a_i - x_i) \tag{7}$$

where a_i is a random attractive position.

3 Source Seeking Analysis

We evaluated the algorithms on source seeking tasks with and without noise.

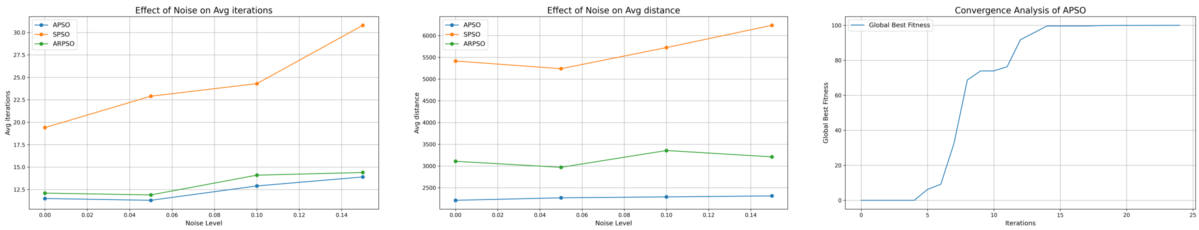
3.1 Source Seeking without Noise



(a) Iterations vs No. of UAVs (b) Distance vs No. of UAVs (c) Convergence Analysis

Figure 1: Source Seeking without Noise

3.2 Source Seeking with Noise



(a) Iterations vs Noise (b) Distance vs Noise (c) Convergence Analysis

Figure 2: Source Seeking with Noise (30 UAVs)

3.3 Conclusions on Source Seeking

- APSO consistently outperforms SPSO and ARPSO across all noise levels, requiring fewer iterations and traveling less distance.
- APSO shows the best scaling as UAV count increases.
- All algorithms degrade as noise increases, but APSO shows the best robustness (least steep slope).
- APSO's acceleration-based approach allows it to maintain momentum through noisy measurements, providing smoother convergence.

4 Optimization: CEC 2022 Benchmark

We utilized the CEC 2022 benchmark suite [1] to rigorously evaluate the algorithms. The suite includes 12 diverse optimization problems (Unimodal, Multimodal, Composition, etc.).

4.1 Hyperparameter Tuning

To ensure fair comparison, we employed Bayesian Optimization to tune hyperparameters for APSO and SPSO. This process builds a probabilistic performance map (Gaussian Process) to balance exploration and exploitation in the hyperparameter space.

- **APSO Tuned:** $w_1 = 0.8, w_2 = -0.285, c_1 = 2.0, c_2 = 1.193$
- **SPSO Tuned:** $w = 0.6, c_1 = 1.5, c_2 = 1.5$

4.2 Results and Observations

4.2.1 2D Results (Low-Dimensional)

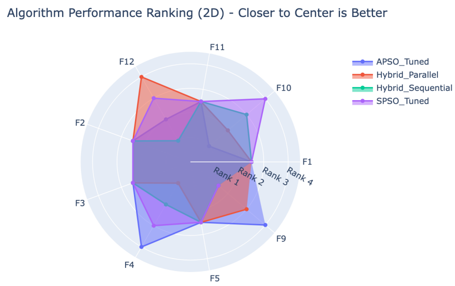


Figure 3: Radar Chart for 2D Results

In 2D, performance is competitive with no single dominant algorithm. Low-dimensionality appears to be a "solved problem" for these PSO variants, with marginal differences.

4.2.2 10D Results (Moderate-Dimensional)

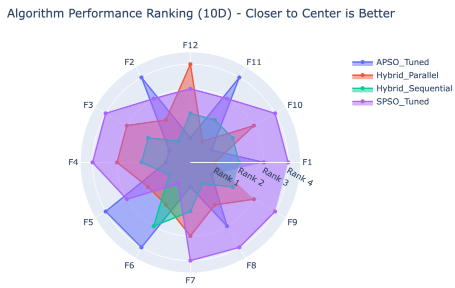


Figure 4: Radar Chart for 10D Results

A distinct performance gap emerges in 10D. APSO is the standout winner (Rank 1 on 10/12 functions). Its second-order dynamics enable effective navigation of complex landscapes where SPSO struggles.

4.2.3 20D Results (High-Dimensional)

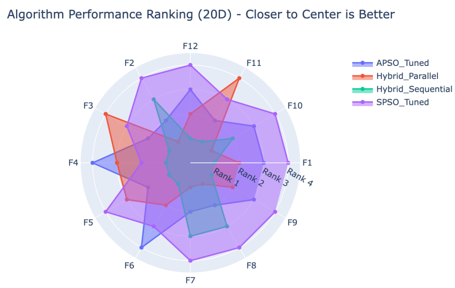
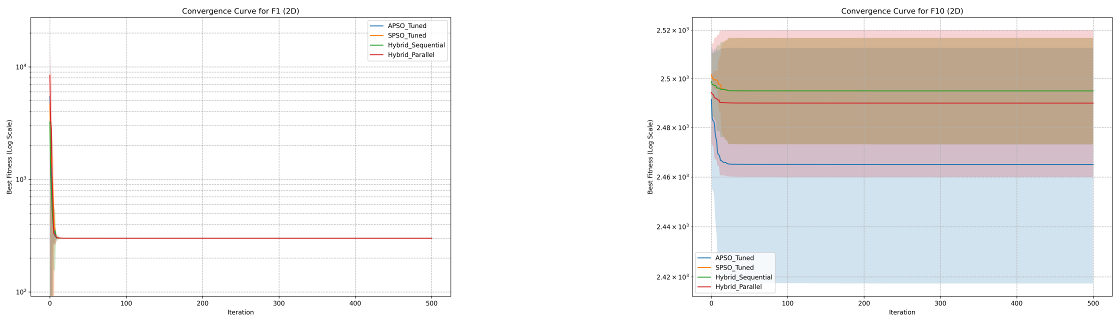


Figure 5: Radar Chart for 20D Results

In 20D, the landscape changes completely. Pure APSO performs poorly, and SPSO ranks last. The Hybrid Sequential strategy emerges as the dominant winner.

4.3 Convergence Analysis

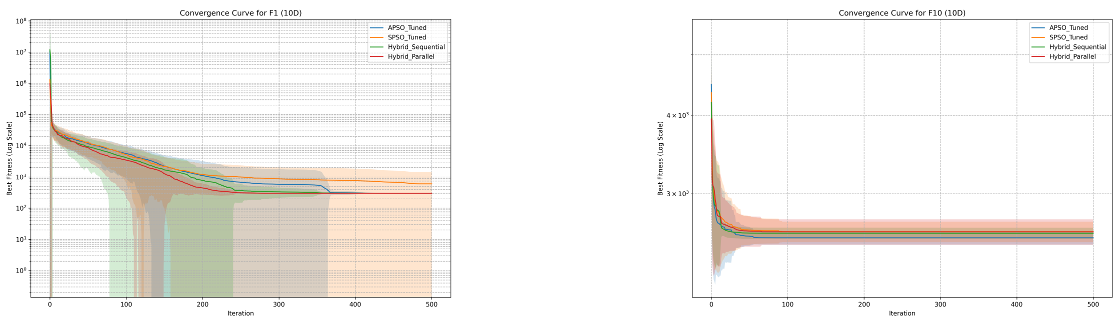
To further understand the optimization behavior, we analyze the convergence plots for representative functions.



(a) F1 (2D)

(b) F10 (2D)

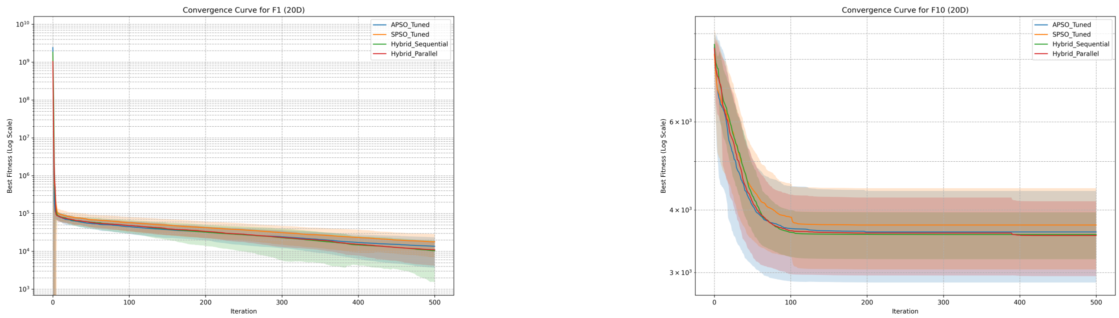
Figure 6: Convergence Plots for 2D



(a) F1 (10D)

(b) F10 (10D)

Figure 7: Convergence Plots for 10D



(a) F1 (20D)

(b) F10 (20D)

Figure 8: Convergence Plots for 20D

The convergence plots illustrate the distinct behaviors of the algorithms. In 2D, the convergence is rapid for all algorithms, with minimal differences. In 10D, APSO (and Hybrid Sequential in its APSO phase) converges rapidly. However, in 20D, the Hybrid Sequential strategy demonstrates its power: the initial exploration phase prevents premature convergence, and the switch to exploitation (SPSO phase) drives the solution to a better optimum than either pure strategy could achieve.

5 Hybrid Exploitation Strategies

To address the limitations of individual algorithms, we tested two hybrid strategies:

5.1 Hybrid Parallel PSO

A single swarm with division of labor. 15 "Explorer" particles (APSO) and 15 "Exploiter" particles (SPSO) operate simultaneously, sharing a global best.

5.2 Hybrid Sequential PSO

A phased approach.

- **Phase 1 (Exploration):** All particles use APSO for the first half of iterations to broadly search.
- **Phase 2 (Exploitation):** All particles switch to SPSO for the second half to fine-tune the solution.

6 Further Improvements

While the Hybrid Sequential strategy has shown remarkable success in high-dimensional spaces, there is room for further enhancement through **Adaptive Hybridization**.

Currently, the switch from exploration (APSO) to exploitation (SPSO) occurs at a fixed point (50% of iterations). An adaptive approach would dynamically switch between algorithms based on real-time performance metrics, such as:

- **Evolutionary Speed:** Measuring how quickly the fitness is improving. If improvement stalls, the algorithm could switch strategies.
- **Aggregation Degree:** Measuring the diversity of the swarm. If the swarm becomes too clustered (low diversity) without finding a good solution, the algorithm could switch to an exploration-heavy mode (APSO or ARPSO) to escape local optima.

Such an adaptive mechanism could potentially offer the "best of both worlds" across all dimensions, eliminating the need to manually select a strategy based on problem complexity.

7 Conclusion

- **Hybrid Sequential Superiority:** This strategy is the most robust overall, dominating in high-dimensional (20D) problems. The phased approach of exploration followed by exploitation is key.
- **Dimensionality Matters:** The best algorithm depends on the problem dimension (APSO for 10D, Hybrid Sequential for 20D), supporting the No Free Lunch Theorem.
- **Structured Hybridization:** Structured hybridization (Sequential) proved more effective than simple parallel mixing.
- **Tuning Importance:** Bayesian Optimization tuning was essential for fair and optimal performance comparison.

References

[1] Wenjian Luo, Xin Lin, Changhe Li, Shengxiang Yang, and Yuhui Shi. Benchmark functions for cec 2022 competition on seeking multiple optima in dynamic environments, 2022.

[2] Adithya Shankar, Himanshu, Harikumar Kandath, and J. Senthilnath. Acceleration-based pso for multi-uav source-seeking. In *IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society*, pages 1–6, 2023.